

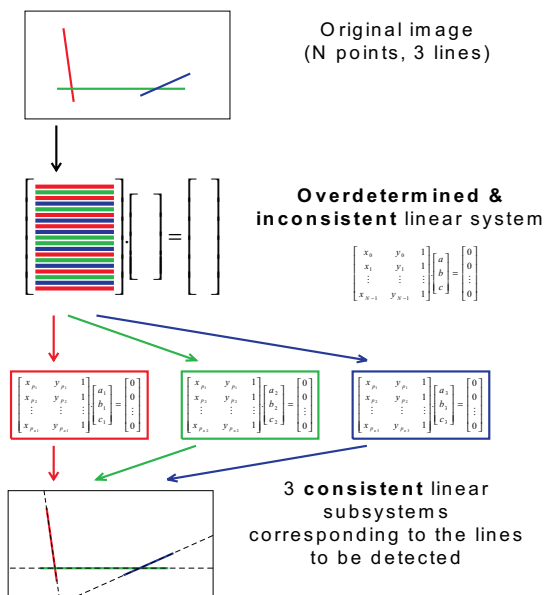
A NEW APPROACH FOR FAST LINE DETECTION BASED ON COMBINATORIAL OPTIMIZATION

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We present a new algorithm for detecting lines in digital images. The algorithm is based on a general combinatorial optimization approach for estimating piecewise linear models. A linear system is constructed with the coordinates of all contour points in the image as coefficients and the line parameters as unknowns. The resulting linear system is then partitioned into a close-to-minimum number of consistent subsystems using a greedy strategy based on a thermal variant of the perceptron algorithm. While the partition into consistent subsystems yields the classification of the corresponding image points into a close-to-minimum number of lines, the solution of each subsystem provides the parameters of those lines.

An extensive comparison with the standard Hough Transform and the Randomized Hough Transform shows the considerable advantages of our combinatorial optimization approach in terms of memory requirements, time complexity, robustness with respect to noise, and quality of the solution regardless of the parameter settings.

The problem of line detection can be formulated as that of partitioning an **inconsistent linear system** into **consistent subsystems**



A thermal variant of the perceptron algorithm

Each iteration $\equiv$ **weighted** projection on a hyperplan corresponding to a point randomly chosen.

Take an arbitrary starting point (initial solution),

Set an initial temperature T_0

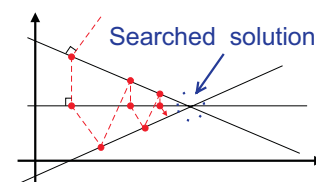
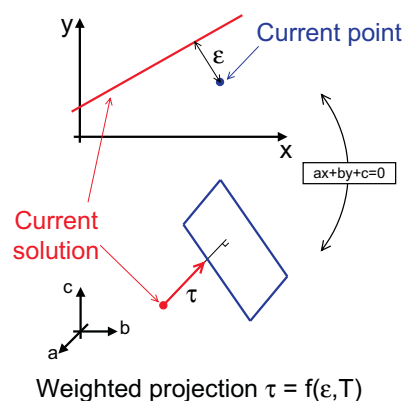
Select a predefined nb of cycles N_0

For each cycle

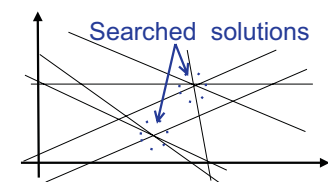
1. Select all equations (ie points) in random order,

2. Update the current solution depending on the temperature and the select equation error,

3. Decrease the temperature (exponential decrease from T_0 to 0).

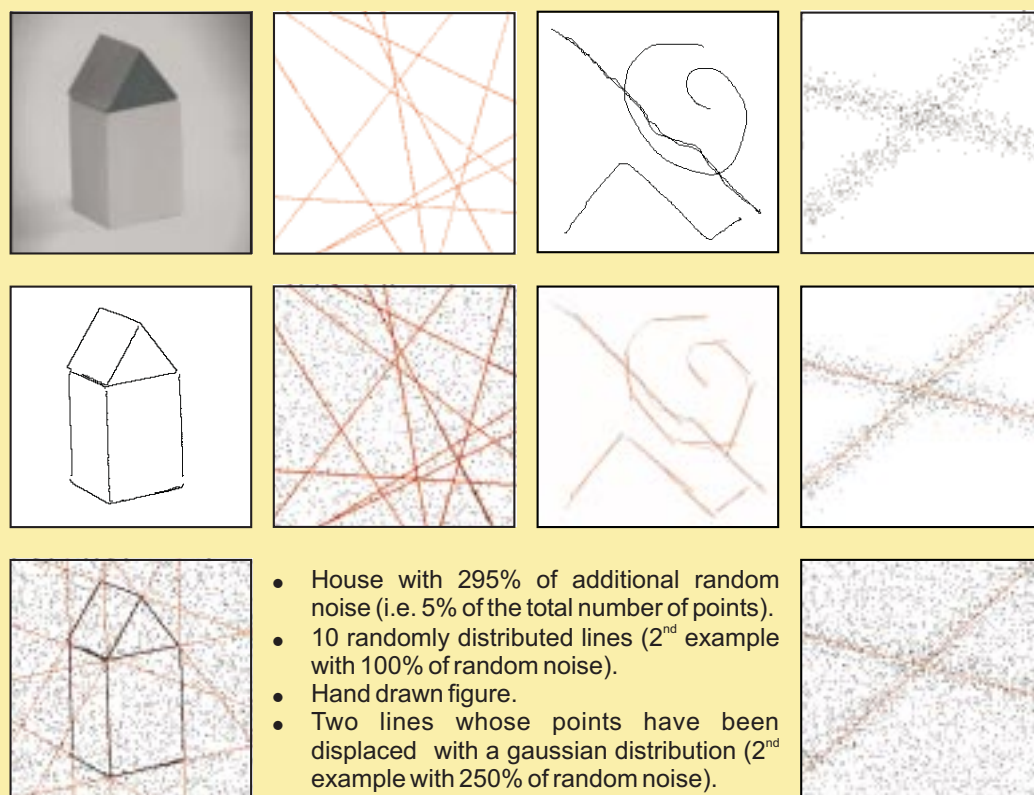


Case of a consistent system (only one line in the image)



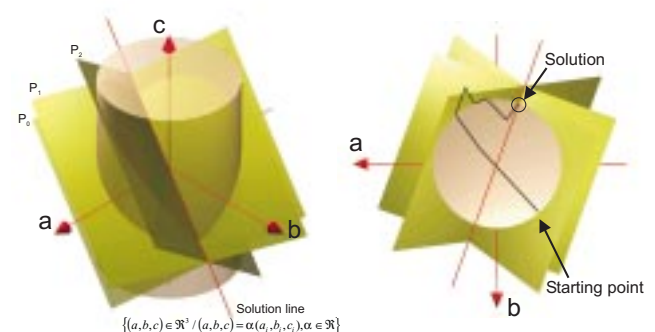
Case of an inconsistent system (several lines in the image)

Simple representation in 2D.
(In reality the reciprocal space is (a, b, c) and each point in the image correspond to a plane in this space.)



A good candidate for :

- Tracking applications or industrial vision system with a priori knowledge. (Very easy to introduce initial solutions or knowledge)
- Extension to higher order (polynomial complexity increase w.r.t. space dimension instead of exponential for the Hough transform.)
- Dedicated implementation (low memory DSP, embedded systems, ...)



3D representation of the case of 3 aligned points $(x_i, y_i)_{i \in [0,1,2]}$

- The system is **consistent**, each point (x_i, y_i) corresponds to a plane P_i in the parameters space ($P_i: ax+by+c=0$).
- The temperature scheme is not represented here (equivalent to $T=\infty$).
- The current solution (a, b, c) is alternatively projected onto a plane p_i (randomly selected) and the unit cylinder.
- The algorithm can converge to two equivalent solutions (intersection points of the cylinder and the solution line) depending on the starting point (initial solution) and the projection order.

